Snowflake SnowPro Core Cheat Sheet

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Exam Information

Average Salary	$90.501
Duration	115 minutes
Exam Guide	Link to the guide.
Format	Multiple Select, Multiple Choice, Interactive questions
Language	English, Japanese, & Korean
Number of Questions	100
Passing Score	750 + Scaled Scoring from 0-1000
Price	175 USD
Recommended course	Link to the course.
Validity	2 years
Version	COF-C02

What is Snowflake?

Snowflake is a Data Solution provided as Software-as-a-Service (SaaS). **It's not available on-premise**. It combines a new SQL query engine with an innovative architecture natively designed for the cloud.

Use cases

Snowflake is optimal for Data Warehouse, Data Lake, Data Exchange, Data Apps, Data Science, and Data Engineering.

Cloud Providers

- Amazon Web Services
- Azure
- Google Cloud Platform

You can have data in Azure and load it into Snowflake on AWS with no problem. Government regions are only supported on Business-Critical Edition or higher on Azure and AWS

Cloud Providers that Snowflake Supports.

Ways to connect to Snowflake

- Snowsight (Web Interface)
- SnowSQL (CLI Client)
- ODBC and JDBC
- SDKs (Node.js, Python, Kafka, Spark, Go, and more.)

Snowflake Editions

- Standard
- Enterprise
- Business Critical
- Virtual Private Snowflake

Virtual Private Snowflake is Like the Business Critical Edition but in a Snowflake environment isolated from all the other Snowflake accounts

Edition	Time-Travel	Multi-Cluster warehouses	Fail-Safe	Data Sharing	Dedicated Virtual Warehouses	Materialized Views	Replication
Standard	0-1 days	No	Yes	Yes	Yes	No	Yes
Enterprise	0-90 days *1 by default*	Yes	Yes	Yes	Yes	Yes	Yes
Business Critical	0-90 days *1 by default*	Yes	Yes	Yes	Yes	Yes	Yes

Edition	Column-Level Security	Query Statement Encryption	Failover / Failback	Federated Authentication	PrivateLink	Tri-Secret Secure Encryption
Standard	No	No	No	Yes	No	No
Enterprise	Yes	No	No	Yes	No	No
Business Critical	Yes	Yes	Yes	Yes	Yes	Yes

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Different Types of Costs

Storage Costs	Average amount of storage used per month, after compression
Compute Costs	Billed in seconds with a one-minute minimum
Cloud Service	Up to 10% of daily compute credits are included for free
Data Transfer	Move or copy their data between regions or cloud providers

The Snowflake edition, warehouse size, number of clusters, and time that each server in each cluster runs determine the number of credits a data warehouse consumes.

Capacity Options

On-Demand	Fixed rate for the consumed services
Pre-paid	Cheaper, but commitment to Snowflake

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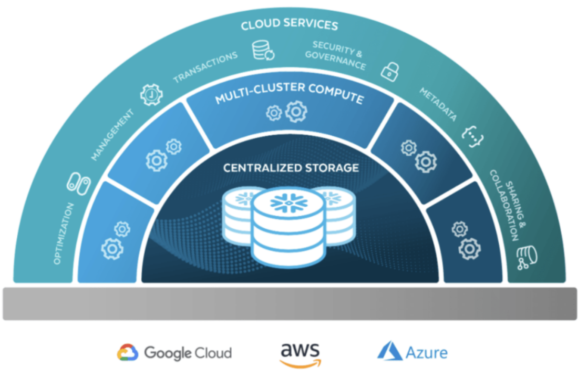
Architecture

Shared-disk architectures	Central data repository for persisted data accessible from all compute nodes in the platform.
Shared-nothing architectures	Each node in the virtual warehouse cluster stores a portion of the entire data set locally.

Hybrid of traditional shared-disk and shared-nothing database architectures.

Architecture Layers

Centralized Storage Layer	Snowflake reorganizes that data into its internal optimized, compressed columnar format.
Compute Layer	Query execution is performed using virtual warehouses in the compute/processing layer
Cloud Services Layer	Collection of services that coordinate activities across Snowflake, including Authentication, Access Control, etc.
Cloud Agnostic Layer	It is only used the first time when we choose a provider.



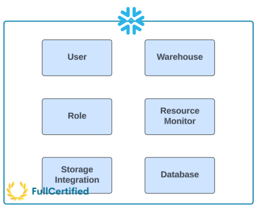
Snowflake Layers.

Snowflake Objects

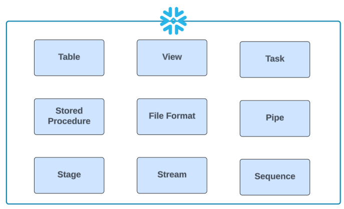
Account	Must be unique.
Warehouse	Virtual Machine to execute queries. Compute Part.
Database	Logical Collection of Schemas.
Schema	Logical Collection of Objects. The Public schema and the Information_Schema are created when creating a Database.

Main Objects contained within a Schema

Tables	Contains all the data in the DB.
Views	Virtual table defined by a query.
Stages	The location of data files in the Cloud Storage
File Formats	Describes a set of staged data to access or load into Snowflake tables.
Sequences	Like counters to create unique numbers.
Pipes	Object that enables automatic loading of data from files as soon as they are available in a Stage.
Stored Procedures & User-Defined Functions (UDF)	Extend the system to perform operations in different programming languages.



Account Level Objects



Schema Level Objects

Micro-partitions

All data in Snowflake tables are automatically divided into micro-partitions, contiguous units of storage between 50 and 500MB of uncompressed data, organized in a columnar way. They are immutable, meaning they cannot be changed once created.

Pruning process

Technique to analyze the smallest number of micro-partitions to solve a query. It retrieves all the necessary data to give a solution without looking at all the micro-partitions, saving a lot of time to return the result.

Load Data

Bulk Load	Loading batches of data from files already available at any stage into Snowflake tables
Continuous Load	Load small volumes of data (micro-batches) and incrementally make them available for analysis.

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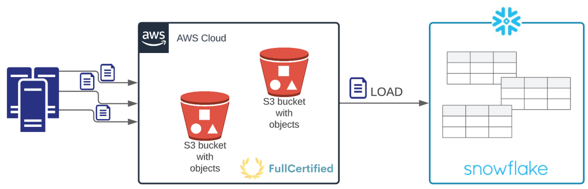
Bulk Load

COPY INTO	Load data from any stage to an existing table. 64 days of metadata.

Some important considerations:
1) You cannot Load/Unload files from your Local Drive
2) Using Snowsight (Snowflake web interface), you can only Load 250MB files
3) Organizing input data by granular path can improve load performance
4) FORCE=True to copy the files again and omit the 64 days of metadata.
5) PURGE = True removes the data files from the stage.
5) If there is any error, you can specify different options: ABORT_STATEMENT, CONTINUE, SKIP_FILE, SKIP_FILE_num, SKIP_FILE_num%.



Continuous Load

Snowpipe	Loading data when the files are available in any (internal/external) stage. 14 days of metadata.
Snowflake Connector for Kafka	Reads data from Apache Kafka topics and loads the data into a Snowflake table
Third-Party Data Integration Tools	You can see the list at the following link.

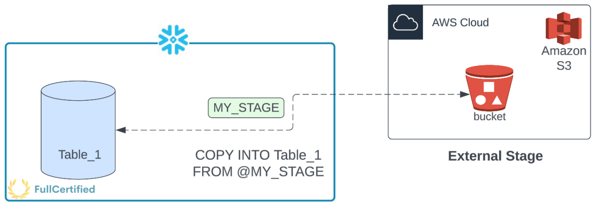
The most important part of this section is Snowpipe. You should use it for **small volume of frequent data, and you load it continuously** (micro-batches). It's **serverless**, which means that it doesn't use Virtual Warehouses.

It can detect new files by automating Snowpipe using cloud messaging or by calling the Snowpipe REST endpoints.

Stages

- Named External Stage
- Named Internal Stage
- User Internal Stage (@~)
- Table Internal Stage (@%)

Storage Integrations will enable users to avoid supplying credentials when creating stages or loading or unloading data. It's an object that stores a generated identity and access management (IAM) entity for your external cloud storage.



Stage Metadata

METADATA$FILENAME	Name of the staged data file the current row belongs to.
METADATA$FILE_ROW_NUMBER	Row number for each record in the container staged data file.

Other commands

PUT	UPLOAD files from a local directory/folder on a client machine into internal stages.
GET	DOWNLOAD files from a Snowflake internal stage into a directory/folder on a client machine
Example	*GET @my_int_stage file:///tmp/data/;*

You cannot use both of these commands with the Snowflake Web UI.

Types of tables

Permanent Tables	Default type of table. Up to 90 days of Time Travel.
Transient Tables	Ideal for temporary data that needs to persist beyond a single session.
Temporary Tables	Store non-permanent data and only exist within the session in which they were created
External Tables	Store data outside of Snowflake.
Iceberg Tables	Manage large datasets stored in external cloud storage efficiently. They use External Volumes, Iceberg Catalog, and Catalog Integration.
Hybrid Tables	Hybrid workloads of OLAP and OLTP operations.

Type of table	Time-Travel Duration?	Is Fail-Safe allowed?	Cloning to other types of tables?	Can views be created?
Permanent	0-90 days	Yes	To Permanent To Transient To Temporary	Yes
Transient	0-1 day	No	To Transient To Temporary	Yes
Temporary	0-1 day	No	To Transient To Temporary	Yes
External	Not allowed	No	No	Yes

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Types of views

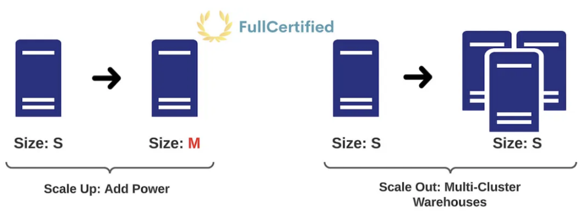
Regular Views	They don't store query results. Useful for frequently changing data where performance is less of a concern.
Materialized Views	They store query results allowing for faster access, but incurring in additional costs.
Secure Views	They ensure that only authorized users can see the view's definition and details, providing an additional layer of security.

	Edition where you can create them	Do they store data?	Is Clustering allowed?	Can they be shared?	Do they support Streams?
Regular View	All Editions	No	No	No	Yes
Materialized View	Enterprise and above	Yes	Yes	No	No
Secure View	Regular: All Editions. Materialized: Enteprise and above	Regular: No. Materialized: Yes.	Regular: No. Materialized: Yes.	Yes	Yes

Data Warehouses Properties

Size	Impact the amount of time required to execute queries
Multi-Cluster Warehouses	Scale compute resources to manage query concurrency
Multi-Cluster Warehouses Modes	Maximized & Auto-scale
Scaling	Scale up/down to increase performance. Scale out/in to improve concurrency for users/queries.
Scaling policies	Standard & Economy
Auto Suspend & Auto Resume	Enabled by default.

A Data Warehouse is a cluster of computing resources in Snowflake that provides CPU, memory, and temporary storage to perform queries and DML operations. While a warehouse is running, it consumes Snowflake credits. It utilizes per-second billing (with a 60-second minimum each time the warehouse starts).



Cache Strategies

Metadata Cache	Objects Information & Statistics.
Warehouse Cache	Attached SSD storage to a warehouse. Information is lost when the warehouse is suspended.
Query Result Cache	It stores the results of our queries for 24 hours. If we perform the same query and the data hasn't changed, it will return the same result without using the warehouse.

You can find a complete example of how to use the different cache strategies at the following link.

Resource Monitors

Credit Quota	Snowflake credits allocated to the monitor for the specified frequency interval
Monitor Level	Monitor the credit usage for your entire Account or individual Warehouses
Schedule	When the monitor is going to start monitoring
Actions (when the threshold is reached)	Notify (send notification), Notify & Suspend (suspend warehouse), or Notify & Suspend Immediately (kill query).

Resource monitors help you control costs and avoid unexpected credit usage caused by running data warehouses. You can impose limits on the number of credits that warehouses consume. They can only be created by AccountAdmins.

Time Travel

Use Cases	Access historical data at any point within a defined period.
Objects we can restore	Databases, Schemas, and tables.
Retention Period	1 day by default, with a maximum of 90 days (Enterprise edition).
Ways to restore	By offset, query statement ID, or timestamp.
Example	*UNDROP TABLE mytable;*

Note: Time Travel requires additional storage, which will be reflected in your monthly storage charges

Fail-Safe

Use Cases	It ensures historical data is protected in the event of a system failure or other catastrophic event
Retention Period	NON-CONFIGURABLE 7-day period
Example	No, you cannot recover this data alone; you **MUST ask Snowflake support**

Note: Fail-Safe requires additional storage, which will be reflected in your monthly storage charges

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Zero-Copy Cloning

Use Cases	Create a snapshot of any table, schema, or Database
Cost	FREE, it doesn't consume storage. It does NOT duplicate data; it duplicates the metadata of the micro-partitions.
Other considerations	Privileges are not cloned. Data History is not cloned.

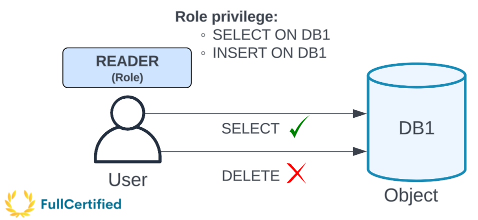
Note: Modifying some cloned data will consume storage because Snowflake has to recreate the micro-partitions, which will cost money.

Access Management Approaches

Discretionary Access Control (DAC)	Each object has an owner who can, in turn, grant access to that object
Role-Based Access Control (RBAC)	Access privileges are assigned to roles, which are, in turn, given to users

Access Management in Snowflake

User	Person or program
Role	Entity to which we grant privileges
Securable Object	Entity to which we can grant access
Privilege	Defined level of access to an object



Default Roles

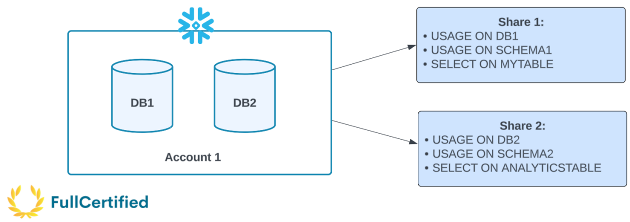
ACCOUNTADMIN	Top-level role
SECURITYADMIN	Manage users and roles
SYSADMIN	Create warehouses and databases (and other objects)
USERADMIN	User and role management
PUBLIC	Automatically granted to every user and role
CUSTOM	Create your own roles and assign the privileges that you want
ORGADMIN	Manages operations at the organization level, i.e., create or view accounts

Role	Have access to the "Account" tab?	Notifications	Can create Shares?	Can manage Network Policies?	Main use-case
ACCOUNTADMIN	Yes	Yes	Yes	Yes	Top-level role
SECURITYADMIN	Yes	No	No	Yes	Manage users, roles, and network policies
SYSADMIN	No	No	No	No	Manage objects
USERADMIN	No	No	No	No	Manage users and roles
PUBLIC	No	No	No	No	Lowest role granted to every user and role by default
CUSTOM	No	No	No	No	*Depends on the assigned privileges*
ORGADMIN	Manages operations at the organization level, including create accounts, view all the accounts in an organization, etc. ** The ORGADMIN role is normally represented outside the roles hierarchy*				

Data Sharing

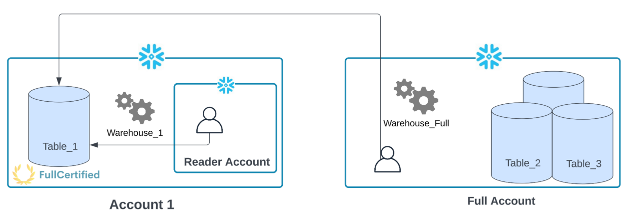
Share	Snowflake objects that encapsulate all information required to share a database
Types of Shares	Outbound & Inbound
Producer	Snowflake account that creates shares and makes them available to other Snowflake accounts to consume
Consumer	Accounts that receive the share/data.

Shared data is instantaneous for consumers as no actual data is copied or transferred between accounts. For this reason, shared data is always up-to-date, and consumers don't pay for storage.



Types of Consumers

Full account	Existing Snowflake account
Reader Account	Someone without a Snowflake account



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Extend Snowflake Functionality

Stored Procedures	Extend Snowflake SQL by combining it with JavaScript
User-Defined Functions (UDFs)	Perform operations that are not available through Snowflake's built-in, system-defined functions. SQL, JavaScript, Java, Python, and Scala. It returns a single row
User-Defined Table Functions (UDTFs)	They can have multiple rows for each input row (the only difference with UDFs)
External Functions	They call code that is executed outside Snowflake

Tasks

Definition	Schedulable scripts that are run inside your Snowflake environment
When they run	Task run on a schedule
Execution	They execute a single SQL statement, including a call to a Stored Procedure
Duration	Maximum duration of 60 minutes by default
Tree of tasks	Each task can have a maximum of 100 children tasks. A tree of tasks can have a maximum of 1000 tasks, including the root one.
Task History	Query the history of task usage within a specified date range
Serverless Tasks	Compute resources automatically scale up or down by Snowflake as required for each workload

Note: Snowflake ensures only one instance of a task with a schedule is executed at a given time. If a task is still running when the next scheduled execution time occurs, that scheduled time is skipped.

Transactions

ACID. Sequence of SQL statements that are committed or rolled back as a unit. Things we need to know for the exam:
1) Snowflake takes 4 hours to abort it if we do not abort it with the SYSTEM$ABORT_TRANSACTION
2) Each transaction has an independent scope.
3) Snowflake does not support Nested Transactions

Streams

Definition	Snowflake objects that record data manipulation language (DML) changes made to tables and views, including INSERTS, UPDATES, and DELETES, as well as metadata about each change
Storage	They don't contain table data; they only store offsets
Types	Standard, Append Only, and Insert Only
Columns	METADATA$ACTION, METADATA$ISUPDATE, METADATA$ROW_ID
	Function that indicates whether a stream contains change data capture (CDC) records

Another important function is *SYSTEM$STREAM_HAS_DATA*, which indicates whether a stream contains change data capture (CDC) records. You can see an example of how streams work at the following link.

File Formats

Structured Data	CSV. The fastest way to load data.
Semi-structured Data	JSON, Parquet, XML, Avro, ORC
FLATTEN	Convert semi-structured data to a relational representation

Format	Type	Load data?	Unload data?	Is it a binary format?
CSV	Structured	Yes	Yes	No
JSON	Semi-structured	Yes	Yes	No
Parquet	Semi-structured	Yes	Yes	Yes
XML	Semi-structured	Yes	No	No
Avro	Semi-structured	Yes	No	Yes
ORC	Semi-structured	Yes	No	Yes

Sequences

Use case	Generate unique numbers across sessions and statements
nextval	Function to generate a set of distinct values

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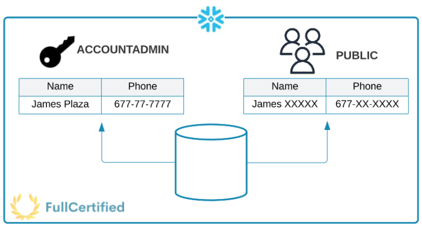
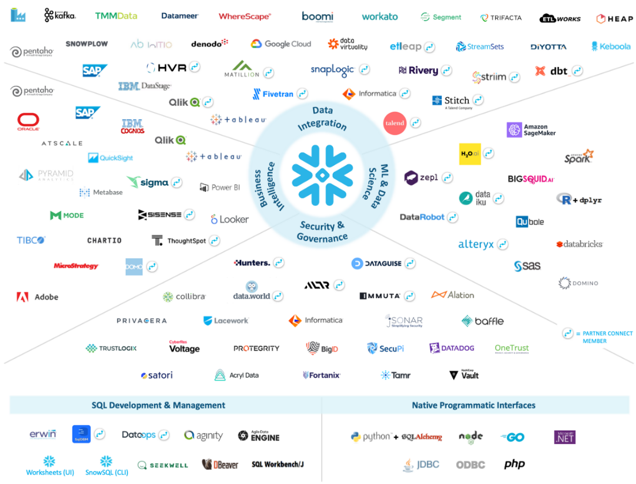
Snowflake Security Features	
Column-Level Security	It includes Dynamic Data Masking and External Tokenization.
Row-Level Security	Control access to specific rows within a table based on certain conditions.
Network Policies	Control and restrict access to your Snowflake account based on IP addresses. ALLOWED IP LIST and BLOCKED IP LIST.
Multi-Factor Authentication (MFA)	Extra layer of security by requiring users to provide two or more verification factors to access Snowflake
Single Sign-On (SSO)	It separates user authentication from user access by using external entities called Identity Providers (IdPs)



Dynamic Data Masking



Other Concepts	
Partner Connect	Technology & Solution Partners
Compliance	HITRUST / HIPAA, ISO/IEC 27001, FedRAMP Moderate, PCI-DSS, etc
Data Marketplace	For providers to buy or sell their datasets. Free, Personalized, and Paid Listings